Windows, Cycles, and Recurrence:The Temporal Geometry of Tangent-Gate Collapse

Allen Proxmire
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Abstract

Collapse in periodic geometries is governed not by continuous interaction but by discrete geometric equivalence. In Event Density Architecture (ED-Arch), a drifting ring encounters its periodic images only at specific times—recurrence windows—when its position modulo the box length returns to a previous equivalence class. Law III shows that these recurrence windows form the temporal scaffold of tangent-gate collapse. Collapse can occur only during these windows, which occur at integer multiples of the recurrence time $T_r = L/v_{\text{drift}}$. This discreteness produces the quantized timing law of Law I: the ladder of collapse times $\chi = \chi_0 + kT_r$.

The width of each recurrence window is determined by the angular tolerance for merge alignment and the drift-phase velocity. Because drift is rapid in the tangent gate, recurrence windows are narrow, explaining the sharp transitions between ladder rungs and the sensitivity of collapse to small changes in particle diameter d. Law III further shows how recurrence interacts with the rational resonance structure of Law II: constructive alignment produces ladder rungs, while destructive alignment produces DECAY islands.

These results establish recurrence as a fundamental law structure of ED-Arch. Recurrence windows generate the ladder, shape the broken ridgeline, and govern the interaction between timing and resonance. Law III therefore provides the geometric foundation on which the remaining laws—N-scaling and commensurability—will build.

1. Introduction

Event Density Architecture (ED-Arch) reveals that collapse in periodic geometries is governed by discrete geometric structures. Law I established the quantized timing law of the tangent gate—the ladder of collapse times indexed by integer recurrence cycles. Law II showed that this ladder is periodically interrupted by DECAY islands, arising from rational resonance between ring circumference and box geometry. Together, these laws describe the timing and resonance structure of tangent-gate collapse.

Yet one question remains unanswered:

Why is collapse time quantized at all?

The ladder is not a numerical artifact, nor a property of the collapse mechanism itself. It arises from a deeper geometric structure: the recurrence geometry of periodic boundary conditions (PBCs). In a PBC domain, a drifting ring does not continuously "see" its periodic images. Instead, it encounters them at discrete, predictable intervals—recurrence windows—determined by the drift speed, box size, and ring geometry.

Law III derives this recurrence structure and shows that:

- collapse can occur only during discrete recurrence windows
- these windows are evenly spaced in time
- their spacing is the ladder step size Δ
- their width determines the sensitivity of collapse
- their interaction with drift phase produces the ladder rungs

In other words:

The ladder exists because PBC recurrence is discrete.

Law III provides the analytic foundation for the quantized timing law of Law I and the resonance structure of Law II. It reveals that the tangent gate is governed not by continuous geometry, but by the discrete recurrence structure imposed by periodicity.

1.1 The Role of Recurrence in Collapse

In the tangent gate, collapse requires that a drifting particle and its PBC image become sufficiently aligned to merge. This alignment does not occur continuously. It occurs only when the drift phase brings the ring into geometric equivalence with its periodic copies.

These equivalence events—recurrences—occur at discrete times:

$$t = kT_r,$$

where T_r is the recurrence time and k is an integer.

Collapse can occur only during these windows.
If the ring misses the window, collapse is delayed until the next one.

This is the geometric origin of the ladder.

1.2 Purpose of Law III

Law III provides the first analytic derivation of:

- the recurrence time T_r
- the ladder step size Δ
- the width of recurrence windows
- the conditions for constructive vs. destructive alignment
- the interaction between recurrence and resonance

It explains:

- why collapse times fall on discrete rungs
- why the rungs are evenly spaced
- why the tangent gate is sensitive to small changes in d
- why DECAY interrupts the ladder

Law III is the geometric backbone of the tangent gate.

1.3 Relationship to Law I and Law II

- Law I described the ladder as a quantized timing law.
- Law II explained the periodic interruptions of the ladder.
- Law III explains the existence of the ladder.

Together:

- Law I = timing
- Law II = resonance
- Law III = recurrence

These three laws form the complete geometric structure of tangent-gate collapse.

2. Recurrence in Periodic Boundary Conditions

Periodic boundary conditions create a discrete equivalence structure. A drifting particle does not continuously interact with its periodic images. Instead, it encounters them at discrete times when its position modulo the box length returns to a previous equivalence class.

These equivalence events are recurrences.

2.1 Definition of Recurrence

A recurrence occurs when the drifting ring satisfies:

$$x(t) \equiv x(t - T_r) \qquad (\mathrm{mod}\, L),$$

meaning the configuration is geometrically identical under PBC wrap-around.

This is not an approximate condition—it is a strict geometric equivalence.

2.2 Recurrence Time T_r

Let v_{drift} be the tangential drift speed of the ring.
A recurrence occurs when the ring has drifted exactly one box length:

$$T_r = L / v_{\mathrm{drift}}.$$

This is the fundamental period of PBC recurrence.

2.3 Recurrence Windows

Collapse requires:

- geometric alignment
- PBC equivalence
- merge-distance proximity

These conditions are satisfied only during narrow windows around:

$$t = kT_r.$$

Thus, collapse is possible only at discrete times.

2.4 Why Recurrence Is Discrete

PBC equivalence classes are discrete.
The ring can only align with its periodic images at integer multiples of T_r.
Between these times, the ring is geometrically incompatible with collapse.

This discreteness is the geometric origin of the ladder.

2.5 Recurrence as the Source of Quantization

Law I observed:

$$\chi = \chi_0 + k\Delta.$$

Law III shows:

$$\Delta = T_r.$$

The ladder is not a numerical artifact.
It is the direct consequence of discrete recurrence geometry.

3. Deriving the Ladder from Recurrence

The ladder structure of the tangent gate—first identified in Law I—is not an emergent pattern or a numerical artifact. It is the direct consequence of discrete recurrence windows imposed by periodic boundary conditions. Collapse can occur only when the drifting ring enters a recurrence window, and these windows occur at integer multiples of the recurrence time T_r. This discreteness produces the quantized collapse times observed in the tangent gate.

In other words:

> **The ladder is the temporal imprint of PBC recurrence.**

3.1 Collapse Requires a Recurrence Window

Let the ring drift tangentially with speed v_{drift}.
A recurrence occurs when the ring has drifted exactly one box length:

$$T_r = L / v_{\text{drift}}.$$

Collapse requires:

- geometric alignment
- PBC equivalence
- merge-distance proximity

These conditions are satisfied only during narrow windows around:

$$t = kT_r.$$

If the ring misses the window, collapse is delayed until the next one.
This is the geometric origin of the discrete collapse times.

3.2 The Ladder Step Size Δ
Law I observed that collapse times follow:

$$\chi = \chi_0 + k\Delta.$$

Law III identifies:

$$\Delta = T_r.$$

The ladder step size is the recurrence time.
This explains why the ladder is:

- evenly spaced
- linear
- independent of collapse mechanism
- sensitive to drift speed

The ladder is not a timing artifact—it is a recurrence artifact.

3.3 Why the Ladder Is Linear
Recurrence windows occur at:

$$T_r,\ 2T_r,\ 3T_r,\ \ldots$$

Because these windows are evenly spaced, the collapse times that fall within them form a linear sequence. The linearity of the ladder is therefore a direct consequence of the uniform spacing of recurrence events.

3.4 Why the Ladder Is Quantized
The ring cannot collapse between recurrence windows.
It is geometrically impossible.

Thus, collapse times must fall on discrete rungs indexed by integer k.
This quantization is not imposed by the collapse mechanism—it is imposed by the geometry of periodicity.

3.5 Why the Ladder Is Universal
Because recurrence geometry is universal across all PBC systems, the ladder structure is not specific to:

- N=3
- the tangent gate
- the collapse mechanism

It is a universal feature of drift-dominated collapse in periodic domains.

4. Geometry of Recurrence Windows

Recurrence windows are not points in time—they are intervals.
Collapse requires that the ring enter a window with sufficiently small angular misalignment. The width of these windows determines the sensitivity of collapse to particle diameter *d*, drift speed, and initial phase.

Law III derives the geometry of these windows and explains why they become narrower with increasing *N*.

4.1 Angular Tolerance

Let $\phi(t)$ be the drift phase of the ring.
Collapse requires:

$$|\phi(t) - 2\pi k| < \epsilon,$$

where ϵ is the angular tolerance for merge-distance alignment.

This tolerance is determined by:

- ring geometry
- particle diameter
- collapse threshold
- PBC image spacing

The narrower the tolerance, the narrower the recurrence window.

4.2 Window Width

The drift phase evolves as:

$$\dot{\phi} = (C(d) / L) \cdot 2\pi.$$

Thus, the width of a recurrence window is:

$$W \approx \epsilon \dot{\phi}.$$

This explains two key empirical facts:

- Larger drift speeds → narrower windows
- Larger circumference → narrower windows

The tangent gate is sensitive because its recurrence windows are narrow.

4.3 Why Windows Narrow with N

For a ring of *N* particles:

$$C(d) = Nd\alpha(N).$$

Thus:

$$\dot{\phi} \propto N.$$

As N increases:

- drift phase advances faster
- recurrence windows shrink
- collapse becomes more sensitive
- the ladder becomes finer

This predicts that the tangent gate becomes increasingly intricate with larger *N*.

4.4 Interaction with DECAY

DECAY occurs when the ring repeatedly misses recurrence windows due to rational misalignment. Law II showed that DECAY islands correspond to values of *d* where:

$$\phi(t) \approx 2\pi p / q.$$

Law III now explains why missing a recurrence window forces collapse to occur at a much later time:

- the ring must wait for the next window
- if misalignment persists, it may miss several windows
- collapse is delayed by integer multiples of T_r

This is the geometric mechanism behind DECAY.

4.5 Recurrence Windows as the Scaffold of the Tangent Gate

The tangent gate is built on three interacting structures:

- recurrence windows (Law III)
- quantized timing (Law I)
- rational resonance (Law II)

Recurrence windows provide the scaffold.
The ladder is the imprint of that scaffold.
DECAY is the resonance-driven interruption of that scaffold.

Together, they form the complete architecture of tangent-gate collapse.

5. Recurrence as a Law Structure

Recurrence is not a computational artifact or a side effect of drift. It is a geometric law structure imposed by periodic boundary conditions. In the tangent gate, collapse is governed by two interacting principles:

- Quantized recurrence (Law III)
- Rational resonance (Law II)

The ladder described in Law I emerges from the first principle.
The DECAY islands described in Law II emerge from the second.

Law III shows that recurrence is the primary scaffold of tangent-gate collapse.
It determines:

- when collapse is possible
- how collapse times are quantized
- why the ladder is linear
- why the tangent gate is sensitive to small changes in d
- how resonance interacts with timing

Recurrence is the geometric backbone of the tangent gate.

5.1 Recurrence as the Generator of Ladder Rungs
The ladder rungs correspond to collapse events that occur during the k-th recurrence window:

$$\chi_k \approx kT_r.$$

This is not an approximation of convenience.
It is a geometric necessity.

Because collapse can occur only during recurrence windows, the ladder rungs are simply the integer multiples of the recurrence time. This explains:

- the linearity of the ladder
- the uniform spacing $\Delta = T_r$
- the independence of rung spacing from collapse mechanism

The ladder is the temporal imprint of recurrence geometry.

5.2 Why Recurrence Windows Are Narrow
The width of a recurrence window is:

$$W \approx \epsilon / \dot{\phi},$$

where ϵ is the angular tolerance for merge alignment and $\dot{\phi}$ is the drift-phase velocity.

Because $\dot{\phi}$ is large in the tangent gate, recurrence windows are narrow.
This explains:

- the sensitivity of collapse to small changes in d
- the sharp transitions between rungs
- the existence of near-miss events (DECAY)

The tangent gate is sensitive because its recurrence windows are thin.

5.3 Constructive vs. Destructive Alignment

Recurrence windows determine whether drift alignment is:

- constructive → collapse occurs (ladder rung)
- destructive → collapse is delayed (DECAY)

Constructive alignment occurs when the ring enters a recurrence window with phase:

$$\phi(t) \approx 2\pi k.$$

Destructive alignment occurs when:

$$\phi(t) \approx 2\pi p / q,$$

with $q>1$, producing the rational misalignments of Law II.

Thus:

- Law III determines when collapse can occur
- Law II determines whether collapse occurs at that time

5.4 Recurrence as the Scaffold of the Tangent Gate

The tangent gate is built on three interacting structures:

- Recurrence windows (Law III)
- Quantized timing (Law I)
- Rational resonance (Law II)

Recurrence windows provide the scaffold.
The ladder is the imprint of that scaffold.
DECAY is the resonance-driven interruption of that scaffold.

This triad forms the complete architecture of tangent-gate collapse.

6. Scaling with *N*

Recurrence geometry is universal across all ring sizes *N*, but its quantitative behavior changes systematically with particle count. Law III extends naturally to larger rings by analyzing how drift speed, circumference, and phase accumulation scale with *N*.

The result is a clear prediction:

the tangent gate becomes more intricate as *N* increases.

6.1 Recurrence Time Scaling

The recurrence time is:

$$T_r = L / v_{\text{drift}}.$$

For a ring of N particles:

- drift speed increases with circumference
- circumference increases linearly with N
- thus T_r decreases with N

This predicts:

- more ladder rungs for larger N
- finer quantization
- denser recurrence structure

The ladder becomes more finely spaced as N grows.

6.2 Window Width Scaling
Window width is:

$$W \approx \epsilon / \dot{\phi}.$$

Since:

$$\dot{\phi} \propto N,$$

we have:

$$W \propto 1 / N.$$

Thus:

- larger $N \rightarrow$ narrower windows
- narrower windows $\rightarrow$ greater sensitivity
- greater sensitivity $\rightarrow$ more DECAY islands

This matches the empirical behavior of the tangent gate.

6.3 Interaction with Resonance Across *N*
As N increases:

- recurrence windows shrink
- resonance spacing shrinks (Law II)
- the two structures interleave more densely

This produces:

- more DECAY islands
- more ladder rungs
- more complex broken-ridge geometry

The tangent gate becomes increasingly resonant and increasingly quantized.

6.4 Universality Across *N*

The recurrence condition:

$$t = kT_r$$

holds for all N.
Thus, recurrence is a universal law structure.

What changes with N is not the existence of recurrence, but:

- the density of windows
- the spacing of rungs
- the sensitivity of collapse
- the interaction with resonance

This universality elevates recurrence from a mechanism to a law of ED-Arch.

7. Discussion

Law III reveals that the ladder structure of the tangent gate is not an emergent pattern or a numerical artifact. It is the direct consequence of discrete recurrence geometry imposed by periodic boundary conditions. In a PBC domain, a drifting ring does not continuously interact with its periodic images. Instead, it encounters them at discrete, predictable times—recurrence windows—determined by drift speed, box size, and ring circumference. These windows form the temporal scaffold on which the ladder is built.

7.1 Recurrence as the Source of Quantization

The quantized timing law of Law I arises because collapse can occur only during recurrence windows. These windows occur at integer multiples of the recurrence time:

$$t = kT_r.$$

Thus, collapse times must fall on discrete rungs indexed by integer k.
The ladder is not a property of the collapse mechanism—it is a property of periodicity.

This explains:

- the linearity of the ladder
- the uniform spacing of rungs
- the independence of rung spacing from collapse details
- the universality of the ladder across parameter sweeps

Quantization is the temporal signature of recurrence.

7.2 Sensitivity and Window Narrowness

Recurrence windows are narrow because the drift-phase velocity $\dot{\phi}$ is large in the tangent gate. This narrowness explains:

- the sharp transitions between rungs
- the sensitivity of collapse to small changes in d
- the existence of near-miss events
- the prevalence of DECAY islands

The tangent gate is sensitive because its recurrence windows are thin.

7.3 Interaction with Resonance (Law II)
Law II showed that DECAY arises when the drift phase satisfies a rational misalignment condition:

$$\phi(t) \approx 2\pi p / q.$$

Law III explains why this misalignment causes collapse to be delayed:

- the ring misses the recurrence window
- collapse cannot occur until the next window
- if misalignment persists, several windows may be missed

Thus:

- Law III determines when collapse is possible
- Law II determines whether collapse occurs at that time

The broken ridgeline of the tangent gate is the superposition of recurrence windows and resonance islands.

7.4 Recurrence as a Universal Structure
Recurrence geometry is universal across all PBC systems.
It does not depend on:

- the collapse mechanism
- the number of particles
- the specific geometry of the ring

It depends only on:

- drift speed
- box size
- periodicity

This universality elevates recurrence from a mechanism to a law structure.

7.5 Relationship to Future Laws
Law III sets the stage for the remaining laws of the ED-Arch series:

- Law IV: N-Scaling of Collapse Surfaces
- Law V: Commensurability Classes

Recurrence provides the temporal scaffold on which these laws will build.

8. Future Work

Law III opens several natural directions for further study. These directions extend the recurrence framework, generalize it across N, and integrate it with the resonance and scaling structures of Laws I and II.

8.1 High-Resolution Mapping of Recurrence Windows

Future work should map recurrence windows for:

- N=3 (baseline)
- N=4,5,6 (scaling regime)

This mapping will reveal:

- window width
- window spacing
- interaction with resonance islands
- sensitivity to initial phase

Such maps will test the predictions of Law III.

8.2 Analytic Derivation of Drift Speed

The recurrence time:

$$T_r = L / v_{\text{drift}}$$

depends on the tangential drift speed, which is currently measured empirically.
A closed-form expression for v_{drift} would allow:

- exact prediction of ladder spacing
- analytic classification of recurrence windows
- extension of Law III to arbitrary N

This is a key step toward a unified analytic theory.

8.3 Recurrence–Resonance Phase Diagrams

Plotting recurrence windows against resonance islands will reveal:

- where constructive alignment occurs
- where destructive alignment occurs
- how DECAY interrupts specific rungs
- how resonance strength varies with k

These diagrams will unify Laws II and III.

8.4 Scaling of Recurrence Geometry with N
Future work should analyze:

- how $T_r(N)$ scales
- how window width $W(N)$ scales
- how ladder density changes with N
- how recurrence interacts with resonance at large N

This will form the foundation of Law IV.

8.5 Toward a Unified Collapse Framework
The ultimate goal is a unified analytic framework that integrates:

- recurrence (Law III)
- resonance (Law II)
- quantized timing (Law I)
- N-scaling (Law IV)
- commensurability classes (Law V)

Such a framework would elevate ED-Arch from an empirical architecture to a predictive mathematical system.